DOMS Page No.
Date / /

ASSIGNMENT NO: 14

Problem Statement: Write a C program to input a binary number from the user and convert it into a decimal number.

Objective:
- To understand no. system conversions.
- To learn how to convert binary nos to decimal numbers in C
- To practice the use of loops & arithmetic operations
- To develop logical thinking for no. system problems

Theory: A binary no. system uses only 2 digits: 0 & 1. Each digit of a binary number represents a powder of 2 depending on its position. To convert a binary no. into its decimal equivalent, each digit is multiplied by 2 raised to power of its position & the results are added together.

Platform: Linux

Input: Binary no. entered by the user

Output: Displays the decimal equivalent of the given binary no.

Sample Input: Enter a binary no: 1011

Sample Output: Decimal equivalent = 11

Conclusion: Thus, the C program successfully converts a binary no. entered by the user into its decimal equivalent using loops & arithmetic operations.

FAQs:

1. What is a binary number system?

A. The binary no. system is a no. system that uses only two digits: 0 & 1. It is the base 2 system and is widely used in computers & digital electronics because electronic circuits can easily represent 2 states [ON = 1, OFF = 0].

Ex: Binary no 1011_2

2. How is a binary no. converted into a decimal no.?

A. To convert a binary no. to decimal:

a) Multiply each binary digit by 2 raised to power of its position

b) Add all the results.

Ex: Convert 1011_2 to decimal

3 2 1 0

$(1 \times 2^0) + (1 \times 2^1) + (0 \times 2^2) + (1 \times 2^3)$

$= 8 + 0 + 2 + 1 = (11)_{10}$

So $(1011)_2 = (11)_{10}$

3) What is the role of loops in Binary to Decimal Conversion?

A. A loop is used to:

- Read each digit of binary no. one by one.
- Calculate its decimal value using powers of 2.

So the loop helps process every bit of binary no automatically.

4) What is the role of the pow() func^n in this program?

A. The pow() func^n is used to calculate powers of nos. In binary to decimal conversion it calculates: $2^{position}$.

Ex: pow(2,0) = 1
pow(2,2) = 4

5) What is the data type used to store binary & decimal nos in this program?

A. Commonly used data types are:
int → to store binary no. & decimal result
int or long → for loop counters or large nos.

Ex:
int binary, decimal = 0;